DATA ANALYTICS / LINKEDIN PROJECT CASE STUDY

HR ANALYTICS & EMPLOYEE ATTRITION

SQL-based workforce analytics and retention-focused business reporting



1,470	16.12%	30.53%
Employees	Attrition rate	Overtime attrition

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Executive Summary

Workforce composition and attrition priorities

1,470	237	1,233
Employee records	Departures	Retained

Project objective

Analyze workforce composition, employee attrition, compensation, satisfaction, experience and performance using SQL. Convert descriptive HR metrics into actionable retention priorities.

Analytical approach

Database preparation and validation; SQL aggregation, grouping, filtering and CASE WHEN segmentation; department-level KPI comparisons.

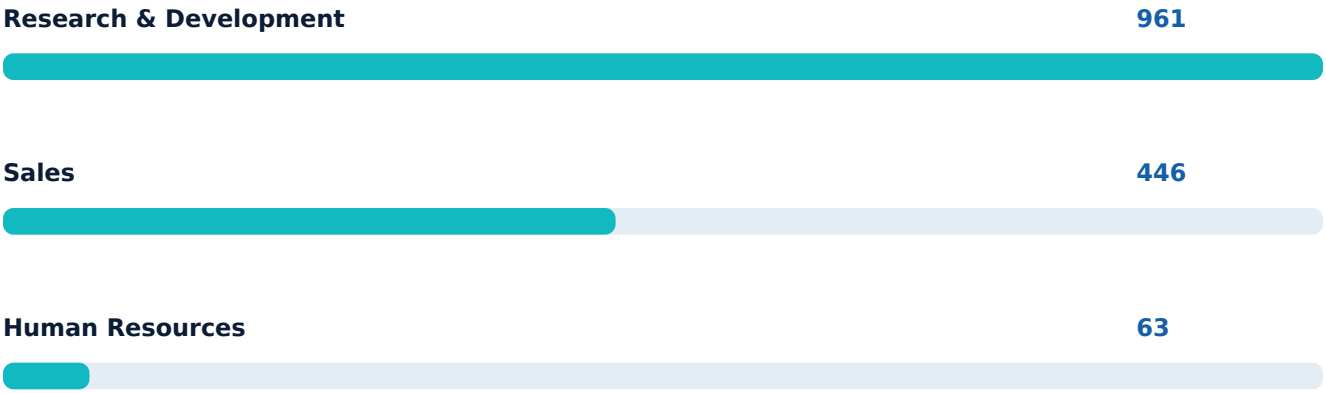
Executive takeaway

Observed attrition is 30.53% among overtime employees versus 10.44% without overtime. Sales has the highest departmental attrition rate at 20.63%. These are associations, not causal findings.

Workforce Profile

Composition, experience and compensation

36.92	$6,502.93	7.01
Average age	Avg. monthly income	Average tenure, years



Gender composition

882 male employees (60.0%) and 588 female employees (40.0%). Average total work experience is 11.28 years.

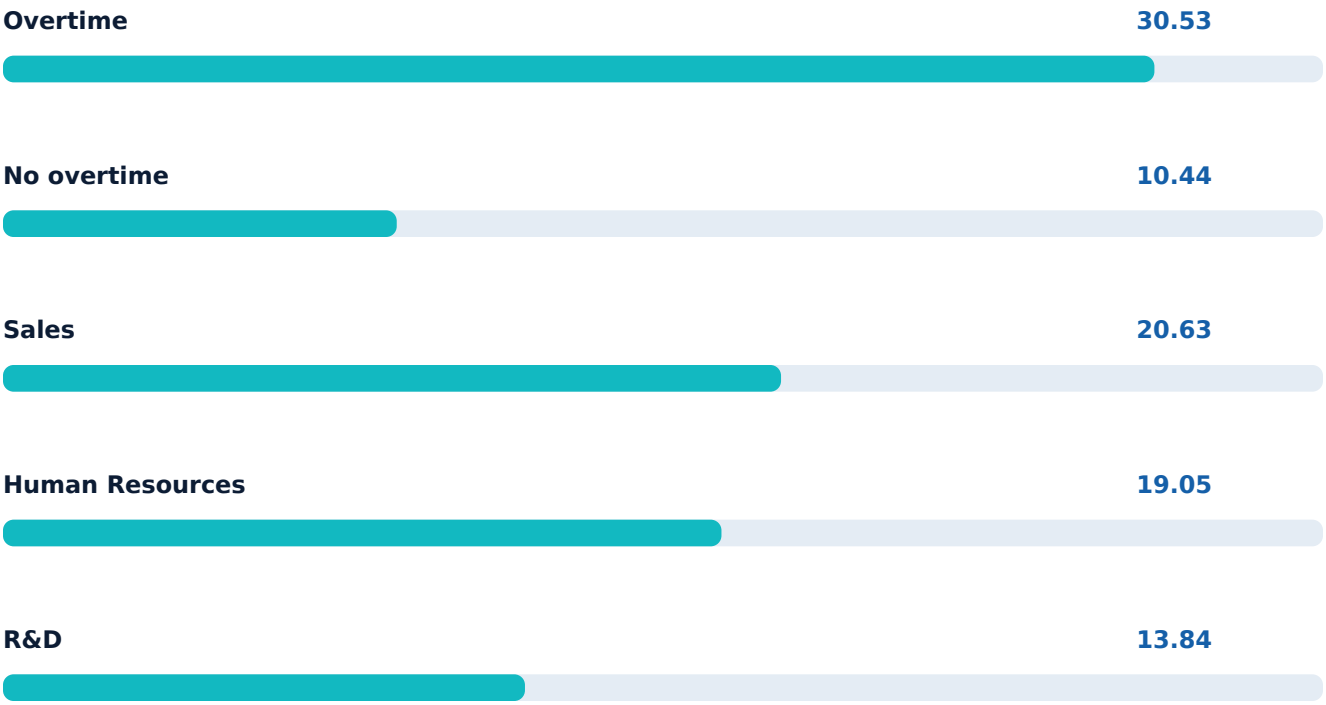
Attrition Diagnostics

Observed differences across employee groups

16.12%
Overall attrition

20.63%
Sales attrition

13.84%
R&D attrition



Overtime employees show approximately 2.9 times the observed attrition rate of non-overtime employees. Other workforce characteristics may explain part of the difference.

Experience and Engagement

Employee segments, ratings and satisfaction

607
Mid-level employees

547
Senior employees

316
Junior employees

Performance ratings

1,244 employees have performance rating 3; 226 employees have rating 4.

Job satisfaction

Reported distribution: level 1 = 289, level 2 = 280, level 3 = 442, level 4 = 459.

Scope of interpretation

The available report does not quantify a direct statistical link between satisfaction, performance ratings and employee departures.

Recommendations and Limitations

Prioritized actions for workforce analytics

- Review workload and overtime exposure, controlling for role and tenure where data permit.
- Prioritize retention analysis in Sales, including exit interviews and manager check-ins.
- Develop mentoring and career pathways for mid-level employees.
- Track engagement, mobility and retention KPIs by department and experience.
- Confirm analysis period, source definitions and underlying SQL before additional modeling.

Source limitation: this portfolio report summarizes descriptive statistics and SQL methods. Raw employee-level data, SQL scripts, significance tests and a reporting period were not supplied with the source PDF.